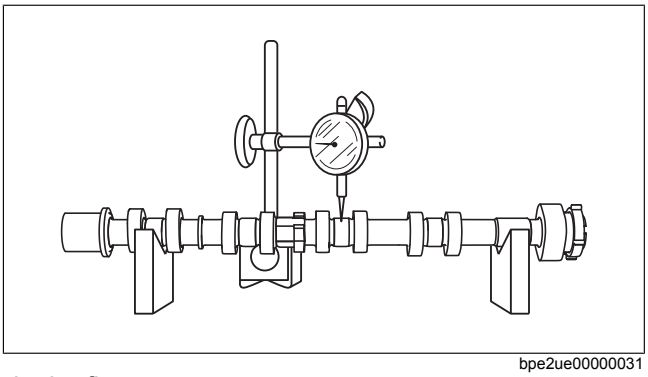
CAMSHAFT INSPECTION

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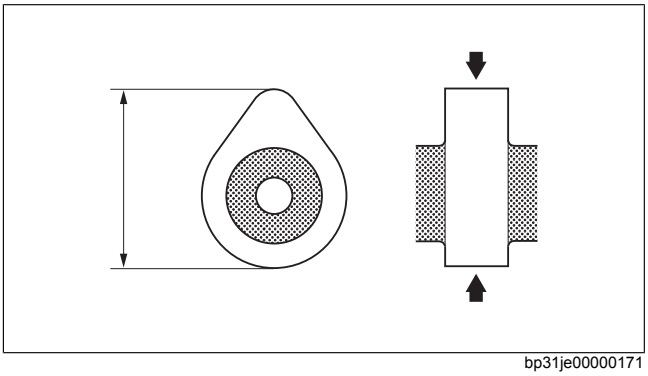
1. Set the No.1 and No.5 journals of the camshaft on V-blocks.
2. Measure the camshaft runout using the dial gauge.
   - If it exceeds the maximum specification, replace the camshaft.

Maximum camshaft runout
0.030 mm {0.0012 in}



bpe2ue00000031

3. Measure the cam height using the micrometer as shown in the figure.
   - If it is less than the minimum specification, replace the camshaft.



bp31je00000171

Standard cam height [SKYACTIV-G 1.3]
IN: 41.54 mm {1.636 in}
EX: 40.34 mm {1.588 in}

Standard cam height [SKYACTIV-G 1.5 (with electric variable valve timing)]
IN: 41.54 mm {1.636 in}
EX: 40.35 mm {1.589 in}

Standard cam height [SKYACTIV-G 1.5 (without electric variable valve timing)]
IN: 41.54 mm {1.636 in}
EX: 40.34 mm {1.588 in}

Standard cam height [SKYACTIV-G 2.0]
IN: 41.57 mm {1.637 in}
EX: 40.37 mm {1.589 in}

Standard cam height [SKYACTIV-G 2.5]
IN: 42.34 mm {1.667 in}
EX: 40.34 mm {1.589 in}

Minimum cam height [SKYACTIV-G 1.3]
IN: 41.47 mm {1.633 in}
EX: 40.27 mm {1.586 in}

Minimum cam height [SKYACTIV-G 1.5 (with electric variable valve timing)]
IN: 41.47 mm {1.633 in}
EX: 40.28 mm {1.586 in}

Minimum cam height [SKYACTIV-G 1.5 (without electric variable valve timing)]
IN: 41.47 mm {1.633 in}
EX: 40.27 mm {1.586 in}

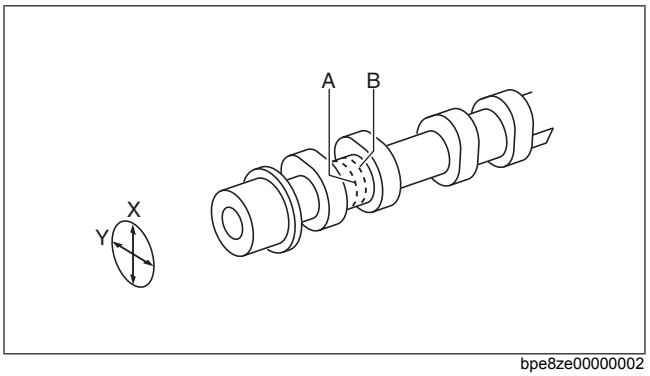
Minimum cam height [SKYACTIV-G 2.0]
IN: 41.50 mm {1.634 in}
EX: 40.30 mm {1.587 in}

Minimum cam height [SKYACTIV-G 2.5]
IN: 42.27 mm {1.664 in}
EX: 40.30 mm {1.587 in}

4. Measure the journal diameter using the micrometer. Measurement positions total four and are in the X and Y directions, at two points (A and B) as shown in the figure.
   - If it is less than the minimum specification, replace the camshaft.

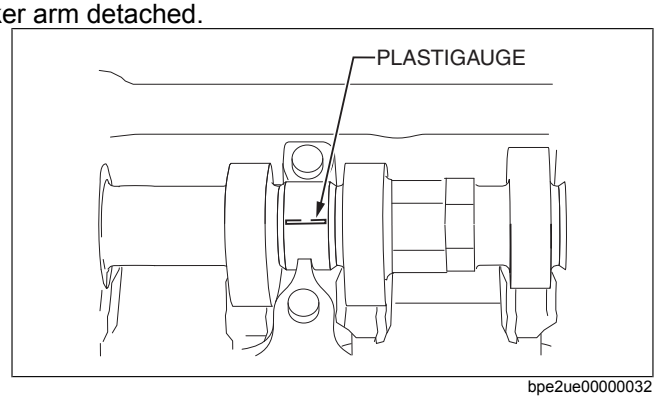
Standard camshaft journal diameter
24.96—24.98 mm {0.9827—0.9834 in}

Minimum camshaft journal diameter
24.93 mm {0.9815 in}



bpe8ze00000002

5. Measure the camshaft journal oil clearance using the following procedure:
   (1) Clean the camshaft journal and the journal receptacle part.
   (2) Put the camshaft on the cylinder head with the rocker arm detached.
   (3) Cut the plastigauge to the same length as the journal width and position it parallel to the camshaft.
   (4) Install the camshaft caps. (See CYLINDER HEAD ASSEMBLY (II).)
   (5) Remove the camshaft caps. (See CYLINDER HEAD DISASSEMBLY (I).)
   (6) Measure the camshaft journal oil clearance.
      • If it exceeds the maximum specification, replace the cylinder head.



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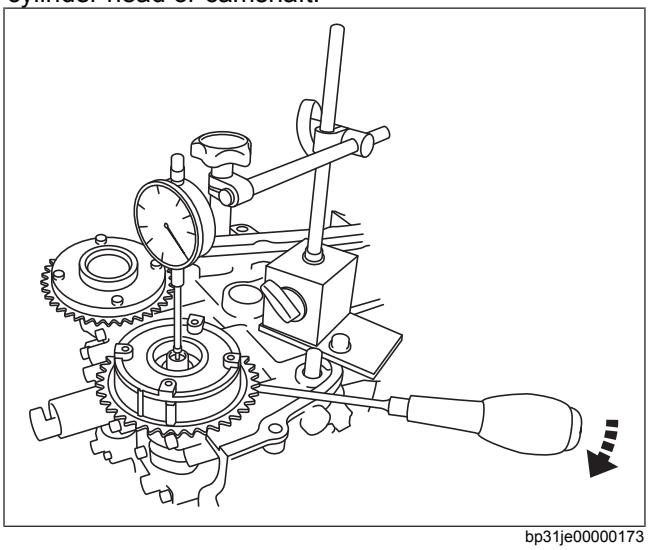
Standard camshaft journal oil clearance
0.035—0.080 mm {0.0014—0.0031 in}

Maximum camshaft journal oil clearance
0.090 mm {0.0035 in}

6. Measure the camshaft end play using a dial gauge.
   • If it exceeds the maximum specification, replace the cylinder head or camshaft.

Standard camshaft end play
0.07—0.22 mm {0.003—0.008 in}

Maximum camshaft end play
0.98 mm {0.039 in}


bp31je00000173